

Interferência de Ruído na Eficiência do Modelo

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Questão 1

Seja

$$H(z) = \frac{1}{z^2 + 0.2z + 0.8} \quad (1)$$

a, b

A figura 1 mostra a resposta ao impulso, obtida pelo trecho de código 1. O comportamento estável, mas oscilatório, condiz com o par de raízes $z = -0.1 \pm 0.89i$, complexas e interiores à circunferência unitária.

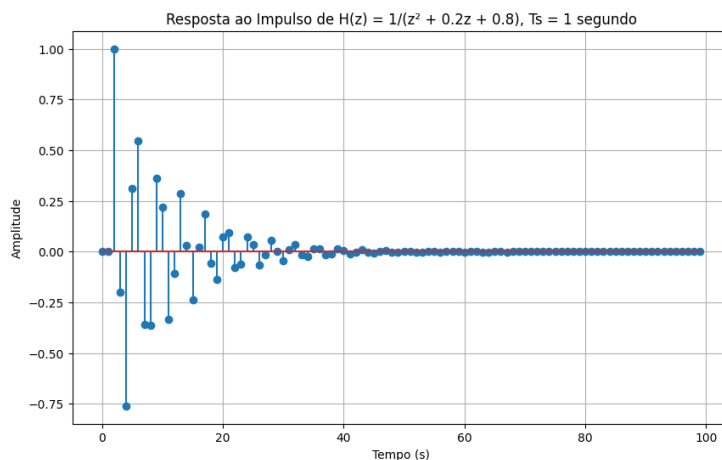


Figura 1: Resposta ao impulso do sistema

c

As figuras ?? e ?? mostram um atraso puro de tempo de 2 segundos, para o tempo considerado $Ts = 1segundo$.

Códigos

Listing 1: Inicializa e simula resposta ao impulso

```
1 import numpy as np
2 from scipy import signal
3
4 import matplotlib.pyplot as plt
5
6 # Define the system in terms of denominator coefficients
7 # For  $H(z) = 1/(z^2 + 0.2z + 0.8)$ , the denominator is [1,
8     ↪ 0.2, 0.8]
9 num = [1] # Numerator coefficients
10 den = [1, 0.2, 0.8] # Denominator coefficients
11
12 # Set the number of samples and the sampling period
13 n_samples = 100
14 Ts = 1 # Sampling period in seconds
15
16 # Compute the impulse response
17 k, impulse_response = signal.dimpulse((num, den, Ts), n=100)
18 impulse_response = np.squeeze(k),
19     ↪ np.squeeze(np.array(impulse_response))
20
21 # Plot the impulse response
22 plt.figure(figsize=(10, 6))
23 plt.stem(k * Ts, impulse_response)
24 plt.title("Resposta ao Impulso de  $H(z) = 1/(z^2 + 0.2z +$ 
25     ↪  $0.8)$ ,  $T_s = 1$  segundo")
26 plt.xlabel("Tempo (s)")
27 plt.ylabel("Amplitude")
28 plt.grid(True)
29 plt.show()
30
31 # Plot the transient response
32 transient_t = k * Ts <= 40 # Focus on the first 40 seconds
33 plt.figure(figsize=(10, 6))
34 plt.stem(k[transient_t], impulse_response[transient_t])
35 plt.title("Resposta Transitoria de  $H(z) = 1/(z^2 + 0.2z +$ 
36     ↪  $0.8)$ ,  $T_s = 1$  segundo")
37 plt.xlabel("Tempo (s)")
38 plt.ylabel("Amplitude")
39 plt.grid(True)
40 plt.show()
41
42 # Plot the start of the response, to focus on the dead time
43 initial_t = k * Ts <= 5 # Focus on the first 5 seconds
44 plt.figure(figsize=(10, 6))
45 plt.stem(k[initial_t], impulse_response[initial_t])
```

```
42 plt.title("Resposta Inicial de  $H(z) = 1/(z^2 + 0.2z + 0.8)$ ,  
    ↳ Ts = 1 segundo")  
43 plt.xlabel("Tempo (s)")  
44 plt.ylabel("Amplitude")  
45 plt.grid(True)  
46 plt.show()
```